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Research Proposal

Currently I am investigating the counting of black hole microstates using the gravitational path integral as part of my undergraduate thesis. This idea is being explored at various levels of supersymmetry. In particular, I am looking at $\mathcal{N} = 8$ and $\mathcal{N} = 4$ supergravity in four dimensions [?, ?].

Throughout these approaches implement the hardy rademacher ramanuhan expansion and due to the conformal symmetry we see the appearance of modular forms.

These beautiful connections between number theory, gravity and string theory have been a source of fascinaition for me. I would like to further explore these connections in a PhD.

Throughout my spare time I have been reading a number of related texts such as Polchinski's String Theory books. This peeked my intrest in Mirror Symmetry and Calabi-Yau manifolds. It is in that context where I discovered the work of Albrecht Klemm.